Closed-Form Mean-Element Theory for GEqOE

The Direct Residue Decomposition Method

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2026

The Problem

Propagating satellite orbits under gravitational perturbations is **expensive**:

- Osculating integration resolves every orbital revolution
- For a LEO satellite: $\sim$16 orbits/day, $\sim$5,800/year
- For catalog maintenance: $\sim$25,000 objects $\times$ 7 days $= 2.8 \times 10^6$ orbit-revolutions

Goal: replace osculating integration with an analytical theory that:

1. Propagates smooth, slowly varying *mean* elements (huge step sizes)
2. Reconstructs the full osculating state at any desired epoch via closed-form corrections

Classical approach: Brouwer (1959) — canonical variables, Lie transforms, closed-form J_2–J_5.

This work: same goal, different elements (GEqOE), different method (direct residue decomposition).

Setup: GEqOE and the Averaging Idea

Generalized Equinoctial Orbital Elements (GEqOE)

State vector (Baù et al. 2021):

$$\mathrm{y} = (\nu,\ p_1,\ p_2,\ K,\ q_1,\ q_2)$$

- ν — generalized mean motion (energy)
- p_1, p_2 — eccentricity vector
- K — generalized eccentric longitude (fast angle)
- q_1, q_2 — inclination vector

Key properties:

- ✓ No singularity at $e = 0$
- ✓ No singularity at $i = 0$
- ✓ No critical-inclination singularity
- ✓ Fast angle K is explicit
- × Singular at $i = 180^\circ$ (retrograde equat.)

Under the unperturbed Kepler problem: $\dot{K} = w/r$ (fast), everything else constant.

Orbit Geometry from K

All geometric quantities are **explicit functions of** K:

$$r = a\big(1 - p_1 \sin K - p_2 \cos K\big) \qquad \text{radius}$$

$$X = a\big(\alpha p_1 p_2 \sin K + (1 - \alpha p_1^2) \cos K - p_2\big) \qquad \text{in-plane } x$$

$$Y = a\big(\alpha p_1 p_2 \cos K + (1 - \alpha p_2^2) \sin K - p_1\big) \qquad \text{in-plane } y$$

where $g = \sqrt{p_1^2 + p_2^2}, \quad \beta = \sqrt{1 - g^2}, \quad \alpha = 1/(1 + \beta)$.

Crucial observation: r, X, Y are all affine in $\sin K$ and $\cos K$.

This means the zonal forcing (which depends on powers of r, X, Y, and $\hat{z} = f(X, Y, q_1, q_2)/r$) is a **trigonometric-rational function of** K.

The Averaging Idea

Separate fast from slow:

$$\underbrace{K}_{\text{fast, once per orbit}} \qquad \underbrace{\nu, p_1, p_2, q_1, q_2}_{\text{slow, secular/long-period}}$$

The **orbital average** over one revolution in K:

$$\langle f \rangle_K = \frac{1}{2\pi a} \int_0^{2\pi} r(K)\, f(K)\, dK = \frac{1}{2\pi} \int_0^{2\pi} f(M)\, dM$$

First-order averaged system (mean elements $\bar{z}$):

$$\dot{\bar{z}} = \varepsilon\, \bar{f}_1(\bar{z}) \qquad \longleftarrow \qquad \text{smooth, slowly varying, large time steps}$$

But the averaged theory needs **two more ingredients**:

1. A map: osculating $\to$ mean (initialization)
2. A map: mean $\to$ osculating (reconstruction)

These are the short-period corrections — and computing them analytically is the hard part.

The Near-Identity Transformation

Mean $\leftrightarrow$ Osculating: The Short-Period Map

Write the near-identity transformation:

$$\mathrm{z}_{\mathrm{osc}} = \bar{\mathrm{z}} + \varepsilon\, \mathrm{u}_1(\bar{\mathrm{z}}, \bar{K}) + O(\varepsilon^2) \qquad \text{(forward map)}$$

$$\bar{\mathrm{z}} = \mathrm{z}_{\mathrm{osc}} - \varepsilon\, \mathrm{u}_1(\mathrm{z}_{\mathrm{osc}}, K_{\mathrm{osc}}) + O(\varepsilon^2) \qquad \text{(inverse map)}$$

The correction u_1 satisfies the **homological equation**:

$$\boxed{\frac{\partial \mathrm{u}_1}{\partial \bar{K}} = \frac{\bar{r}}{\bar{w}} \left[\mathrm{f}_1(\bar{\mathrm{z}}, \bar{K}) - \bar{\mathrm{f}}_1(\bar{\mathrm{z}}) \right]}$$

Translation: u_1 is the antiderivative (in K) of the *oscillatory part* of the forcing, weighted by $r/w = dt/dK$.

The forcing f_1 is a rational-trigonometric function of K.
The weight r/w is also a trigonometric function of K.

$\Rightarrow$ We need to integrate rational-trigonometric functions of K in closed form.

The Integration Challenge

The short-period integrals look like:

$$\mathrm{u}_1(\bar{K}) = \int^{\bar{K}} \frac{r(\phi)}{w} \left[\mathrm{f}_1(\bar{\mathrm{z}}, \phi) - \bar{\mathrm{f}}_1(\bar{\mathrm{z}})\right] d\phi$$

For a zonal potential of degree n:

- f_1 involves $P_n(\hat{z})$ — a polynomial of degree n in $\sin u$
- Times $1/r^{n+1}$ — a power of $(1 - g\cos G)$
- Times r/w — the Jacobian weight

The classical approach: **tangent-half-angle substitution** $t = \tan(G/2)$.

$\Rightarrow$ Rational integrals in t, solvable in principle.

$\Rightarrow$ But the expressions explode in complexity for $n \geq 3$.

Is there a better integration variable?

The Key Insight: Complex True Anomaly

Enter the Complex True Anomaly

Define $F = e^{if}$ where f is the true anomaly.

The Möbius map connects F to $z = e^{iG}$ (eccentric anomaly exponential):

$$F = \frac{z - q}{1 - qz}, \qquad z = \frac{F + q}{1 + qF}, \qquad q = \frac{g}{1 + \beta}$$

On the physical orbit, both z and F trace the unit circle $|F| = 1$ as K advances by 2π.

Three structural advantages of F:

1. The zonal forcing is a Laurent polynomial in F (not a rational-trigonometric function requiring substitutions)
2. The Jacobian dM/dF has a fixed pole structure (independent of the zonal degree n)
3. The short-period integral becomes a contour integral of a rational function

Why F Makes the Forcing Simple

Zonal forcing involves powers of $\sin u = \sin(\omega + f)$:

$$\sin u = \frac{wF - (wF)^{-1}}{2i}, \qquad w = e^{i\omega}$$

This is a Laurent monomial in F (times powers of w).

$\Rightarrow P_n(\hat{z}) = P_n(s_i \sin u)$ is a **finite Laurent polynomial** in F:

$$P_n(s_i \sin u) = \sum_{k=-n}^{n} c_k(Q)\,(wF)^k$$

Combined with $r^{-(n+1)}$ (also rational in F), the complete forcing for each zonal degree n is a **rational function of** F whose numerator is a Laurent polynomial of bounded degree.

Comparison: in K, the same forcing involves nested products of $(1 - g\cos G)^{-k}$ and trigonometric polynomials — much harder to manipulate.

The Jacobian: Fixed Poles

The Jacobian between the mean anomaly M and the complex true anomaly F is:

$$\boxed{\frac{dM}{dF} = -i\,\frac{(1-q^2)^3}{1+q^2}\,\frac{F}{(F+q)^2(1+qF)^2}}$$

Pole structure:

- Simple zero at $F = 0$
- Double pole at $F = -q$ (inside the unit disk, $|-q| < 1$)
- Double pole at $F = -1/q$ (outside the unit disk)

The denominator $(F+q)^2(1+qF)^2$ is **independent of the zonal degree** n.

This is the key structural fact: no matter how high the zonal degree, the integration always has the **same** pole structure. Only the numerator grows with n.

The Direct Residue Decomposition

The Universal Integrand

For any zonal degree n, any slow variable v, and any ω-harmonic m, the short-period integral has the form:

$$u^v_{1,m}(F) = \int \underbrace{a_{v,m}(F)}_{\text{Laurent poly in } F} \cdot \underbrace{\frac{dM}{dF}}_{\text{fixed poles}} \, dF$$

After clearing negative F-powers (shift $s = -k_{\min}$):

$$\boxed{\text{Integrand} = \text{scale} \times \frac{N(F)}{F^s\,(F+q)^2\,(1+qF)^2}}$$

where $N(F)$ is a polynomial in F with coefficients in $\mathbb{Q}(q, Q)$.

This is just a rational function!

$\Rightarrow$ Partial fraction decomposition gives the antiderivative in closed form.

The Five-Step Algorithm

1. **Polynomial long division**
   $N(F) = Q_{\text{poly}}(F) \cdot D(F) + N_{\text{rem}}(F), \quad \deg N_{\text{rem}} < \deg D$
2. **Double-pole residues** at $F = -q$ and $F = -1/q$
   Extract principal parts by evaluation at the poles
3. **Residues at** $F = 0$ (for shift $s \geq 2$)
   Taylor recurrence for the coefficients of $N_{\text{rem}}(F)/[(F+q)^2(1+qF)^2]$
4. **Assemble antiderivative**
   $u^{\vee}_{1,m}(F) = \text{polynomial} - \dfrac{B}{F+q} - \dfrac{D_c}{q(1+qF)} - \sum_j \dfrac{E_j}{F^{j-1}} - C$
5. **Zero-mean gauge**
   Fix the constant C so that $\langle u^{\vee}_{1,m} \rangle_M = 0$ using structural mean formulas

No logarithmic terms appear — guaranteed by the periodicity of u_1 on $|F| = 1$ (single-valuedness forces all log coefficients to vanish).

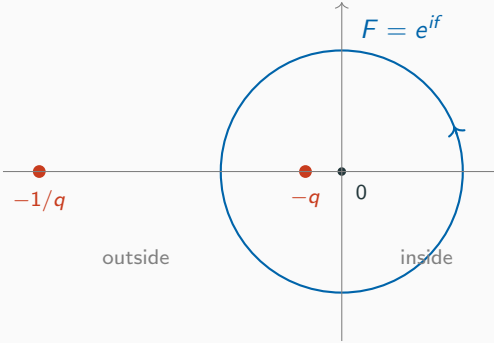
Why Logs Cancel: The Geometric Argument

On the physical orbit, F traces the **unit circle** $|F| = 1$ as M advances by 2π.

The standard rational integration of $1/(F+q)$ would give $\log(F+q)$. But log is multi-valued on a closed contour.

The guarantee: u_1 must be single-valued (periodic in M). A simple pole at $F = -q$ would produce $\alpha \log(F+q)$, which is multi-valued on $|F| = 1$ (the circle passes on both sides of $-q$). $\Rightarrow \alpha = 0$.

The residue theorem confirms: the contour integral of the zero-M-average integrand times dM/dF has poles only at $F = -q$ and $F = 0$ inside the disk. The $F = 0$ residue is



Periodicity of u_1 on the unit circle forces all log coefficients to vanish identically.

Why This Scales

Degree	Forcing terms	Result size	Generation time
J_2	14–21	18 KB	seconds
J_3	32–36	79 KB	$\sim$1 min
J_4	44–55	206 KB	$\sim$15 min
J_5	72–78	510 KB	$\sim$60 min

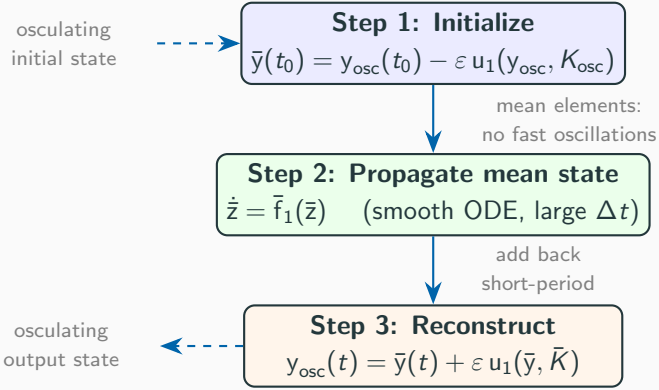
The denominator $(F+q)^2(1+qF)^2$ is degree-independent.

As n increases:

- Only the *numerator* polynomial $N(F)$ grows (linearly in n)
- Algorithm steps (long division, evaluation, recurrence) are polynomial-time in the numerator degree
- The *coefficient complexity* in $\mathbb{Q}(q, Q)$ is the real cost driver (GCD simplification)

The Complete Analytical Theory

The Three-Step Pipeline



Every operation is the evaluation of a **precomputed algebraic formula**. No numerical quadrature, no interpolation, no iterative scheme (except the standard Kepler equation).

For Pure J_2: Fully Closed-Form

Under J_2 alone, the averaged disturbing function depends only on a, e, i.
$\Rightarrow$ No argument-of-pericenter dependence $\Rightarrow$ **uniform rotations**:

$$\begin{bmatrix} \bar{p}_1(t) \\ \bar{p}_2(t) \end{bmatrix} = \begin{bmatrix} \cos\theta_\Psi & \sin\theta_\Psi \\ -\sin\theta_\Psi & \cos\theta_\Psi \end{bmatrix} \begin{bmatrix} \bar{p}_1(t_0) \\ \bar{p}_2(t_0) \end{bmatrix}, \qquad \theta_\Psi = \omega_{\Psi,J_2}(t - t_0)$$

- Magnitudes $\bar{g}$, $\bar{Q}$ constant
- Angles $\bar{\Psi}$, $\bar{\Omega}$ advance linearly
- No ODE integration needed — the mean state is a closed-form function of elapsed time

For general zonals (J_3–J_5): the mean system is a smooth autonomous ODE in $(\bar{g}, \bar{Q}, \bar{\Psi}, \bar{\Omega})$, driven by the single slow angle $\bar{\omega} = \bar{\Psi} - \bar{\Omega}$.

Step sizes: **tens to hundreds of orbital periods**.

Validation: Osculating Reconstruction

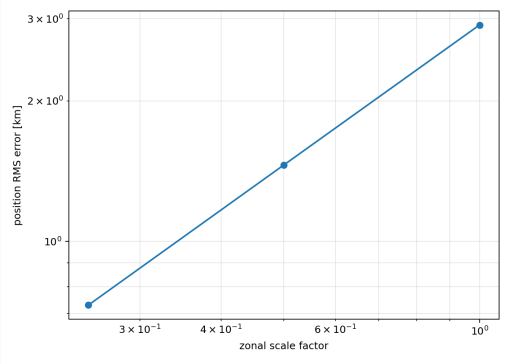
Low eccentricity ($e = 0.05$):

- Position RMS: $\sim$200 m
- Radial error: $\sim$3 m
- 10 orbits, J_2–J_5

High eccentricity ($e = 0.65$):

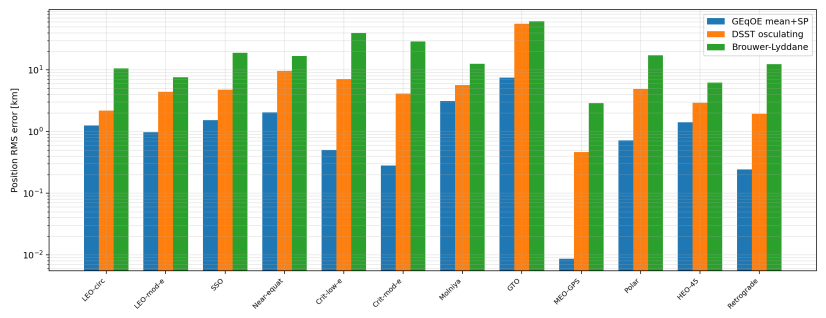
- Position RMS: $\sim$3 km
- Radial error: $\sim$10 m
- 8 orbits, J_2–J_5

Scaling test: log-log slope $= 0.998$ $\Rightarrow$ errors scale as $O(\varepsilon)$ — first-order theory confirmed.



RMS position error vs. zonal scale factor. Slope ≈ 1 confirms first-order behavior.

12-Regime Comparison vs. Brouwer–Lyddane



GEqOE theory outperforms Brouwer–Lyddane (Orekit) by 4× to 335× across all 12 regimes.

Both compared against Cowell Cartesian truth with identical J_2–J_5 constants.

The Big Picture

Why the Residue Decomposition is the Unlocker

GEqOE: r, X, Y affine in $\sin K$, $\cos K$

$F = e^{if}$: forcing $\to$ Laurent polynomial

dM/dF: **fixed poles** $(F+q)^2(1+qF)^2$

Residue decomposition: rational $\to$ closed-form

Closed-form short-period corrections $\mathrm{u}_1(F)$

Closed-form mean drift $\bar{\mathrm{f}}_1(\bar{\mathrm{z}})$

Complete analytical theory

Extensibility: The Same Method Unlocks Everything

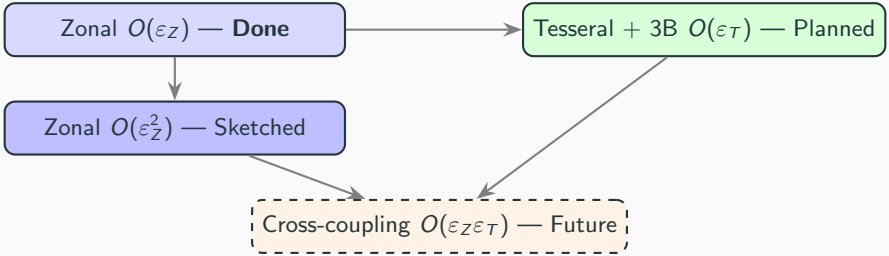
The pole structure $(F+q)^2(1+qF)^2$ is a property of **Keplerian geometry**, not of the perturbation source.

$\Rightarrow$ Any perturbation whose forcing is rational in F can be processed by the same algorithm:

Perturbation	Rational in F?	Mean drift
Zonal harmonics J_n	Yes	Autonomous
Tesseral harmonics C_{nm}, S_{nm}	Yes (frozen θ_G)	Non-autonomous
Third-body (Sun, Moon)	Yes (frozen $\hat{\mathrm{r}}'$)	Non-autonomous
Solar radiation pressure	Yes (frozen $\hat{\mathrm{r}}_\odot$)	Non-autonomous
Fourier-in-K thrust	Yes (by construction)	Controlled

One algorithm, one pole structure, all perturbation sources.

The Road Ahead



- Each layer is **independent** — developed and published separately
- The residue decomposition is the **shared engine** across all layers
- Combined theory: $O(\varepsilon_Z^2) + O(\varepsilon_T) + O(\varepsilon_{3B})$ — captures all operationally relevant terms

Summary

1. **GEqOE elements** provide a non-singular fast angle K with affine orbit geometry
2. The **complex true anomaly** $F = e^{if}$ converts the forcing to Laurent polynomials
3. The **Jacobian** dM/dF has a fixed pole structure independent of the harmonic degree
4. The **direct residue decomposition** converts the short-period integrals to exact closed-form rational expressions
5. The result is a **fully analytical** mean-element theory: precomputed formulas evaluated at runtime, no quadrature
6. The method **generalizes** to tesseral harmonics, third-body, and thrust — same poles, same algorithm
7. Validated: **4–335**$\times$ **better** than Brouwer–Lyddane across 12 orbital regimes

Thank you